

Year 2026
Cohort Kulfi Collective

Team Name Kulfi Krabs
Project Name Stranger Phone
Team Members Khushi Devashree
Email IDs
Team Mentor Kushal Bhor
Team Mentor

Notes

- * Use this template and fill in the details below. Share it back
- * Explain your project in brief

* For all the components that you are ordering online provide t

* For the components that you are planning to purchase from you

* DO NOT RELY ON GETTING ANYTHING FROM THE LOCAL STORE HERE IN COMPONENTS FROM THE LOCAL STORE

Project Brief: A vintage push-button telephone mounted in handset, chooses a language (1 English / 2 language, then presses 4 to hear a message stranger (6 to re-record, 7 to save). Mess only to listeners who chose that language. sticker for a small web 'surprise'. Built onboard keys); powered by a power bank. Pi

Sr no	Product name	Qty
1	Vintage push-button telephone (Bakelite / BSNL style) with handset, cord, keypad and hook	1
2	12 mm tactile push buttons (pack of 10)	1

3	Micro limit switch with lever (2 pcs)	1
4	Ai-Thinker ESP32-A1S Audio Kit v2.2 (ES8388 codec, microSD, 6 keys, headphone + speaker	2
5	microSD card 32 GB Class 10 (HP mi210)	1
6	USB microSD card reader	1
7	MAX9814 microphone amplifier module with electret capsule (AGC)	2
8	Electret microphone capsule 6 mm / 9 mm (pack of 10)	1
9	Resistor assortment kit (1/4 W, 50 values, 500 pcs)	1
10	3.5 mm AUX male-to- male cable (cut in half for two solderable plugs)	1

	11	Mini speaker 8 Ohm 0.5 W, 28 mm (2 pcs)	1
	12	4-core flexible cable, 1 m (or spare coiled handset cord)	1
	13	NFC sticker tags NTAG213 (pack of 10)	1
	14	Power bank 10,000 mAh (EVM EnBolt 22.5W)	1
	15	Micro-USB DATA cable, 1.5 m (2 pcs)	1
	16	Hookup wire 22 AWG, 5 colours x 2 m	1
	17	Female-female Dupont jumper wires, 20 cm (40 pcs)	1
	18	Male pin header strip 2.54 mm (1 pc, 40 pins)	1

19 Heat-shrink tubing assortment	1
20 Hot glue sticks (10) + double-sided foam tape	1
21 Acrylic base plate 200 x 200 x 3 mm + 4x M4 bolts/nuts + cable ties	1
22 Rubber grommet 8-10 mm (2 pcs)	1

Prices and stock checked on 25 Sep 2026 (Probots, Robu, Amazon.
Chor Bazaar bargaining and is the lever if the total runs over.
is being bought in Goa.

TOOLS TO CARRY (owned / borrowed, not purchased - add ~Rs 1,500

- Soldering iron (25-40 W) + solder + flux
- Wire stripper / cutter
- Multimeter (continuity for hook switch + keypad tracing)
- Small screwdriver set (phone screws) + hand drill or drill bi
- Hot glue gun
- Laptop with Arduino IDE, ESP32 board package, arduino-audio-t
- NFC-capable phone with the 'NFC Tools' app (to write the URL

PREP BEFORE GOA (the flowchart cannot run without these - no pu

- Record the prompt files as 16 kHz mono WAV: ask_lang (triling
<language> after the beep'), review (6/7), last_take, saved, su

- Record 5+ seed messages per language into archive/en, archive
- Build and host the surprise web page (he/she/they -> emoji +
- Flash and test the firmware on the bench with the full flow (including the power-bank keep-alive: Wi-Fi radio left on + 47 o
- Pre-drill the base plate; pack the spare board, spare SD card

Angad

link with "editable" link.

the reference link of the online store from where you are
or local store provide the reference link of image or video
GOA. YOU WON'T HAVE THE TIME, MEANS OR AVAILABILITY TO GET THE

inside a Mumbai kaali-peeli cab. A passenger picks up the
(Hindi / 3 Marathi), hears a short description in that
a previous stranger left, or 5 to record one for the next
pages are stored on a microSD card by language and played back
After saving, the passenger taps their phone on an NFC
on an ESP32-A1S Audio Kit (records + plays audio, SD card,
dollars from fieldwork: empathy, connection, time.

Category	Purpose	Reference link
Electro-mechanical	Flow: 'Picks up the phone', keys 1-7, 'Hangs up'. Shell, handset (mic + earpiece), keypad,	https://www.olx.in/items/q-vintage-telephone
Electronics	Flow: keys 1-7. BACKUP: most push-button phones have a printed keypad matrix that cannot	https://robu.in/product/12x12x12mm-tactile-push-button-switch-10pcs/

Electro-mechanical	Flow: 'Picks up' / 'Hangs up'. BACKUP hook switch if the phone's original cradle contacts are	https://robu.in/product/kw1-103-z2a100-spdt-short-lever-micro-switch1n/
Electronics	The brain for every node: plays language/description /menu/record/review/ saved prompts from	https://probots.co.in/esp32-a1s-audio-development-board-kit-ai-thinker-wifi.html
Electronics	Flow: every prompt file (prompts/en, hi, mr) and the message archive (archive/en, hi,	https://www.amazon.in/dp/B07DJLFMPS
Electronics	Loading the 15 prompt WAV files and seed messages onto the SD card from a laptop, and pulling	https://www.amazon.in/dp/B0D9H3W8V6
Electronics	Flow: '(person records message)'. Sits inside the handset mouthpiece; AGC evens out	https://robu.in/product/max9814-high-performance-microphone-agc-amplifier-module-cma-
Electronics	FALLBACK mic: if the MAX9814 signal is too hot or noisy, a bare capsule soldered straight to	https://www.amazon.in/Yellow-Tommy-Condenser-Microphone-Directional/dp/B0GTQZ9ZNK
Electronics	10 kOhm pull-up for key 1 on I036 (input-only pin); 10k/1k divider between MAX9814 and MIC1;	https://www.amazon.in/dp/B0D6LRHHVZ
Electronics	Flow: every 'hears' node. Carries audio OUT of the board's 3.5 mm headphone jack to the handset	https://robu.in/product/aux3-5mm-audio-extension-cable-2m-male-to-male-grey/

Electronics	Replacement earpiece only if the phone's original earpiece is dead. Driven from the headphone jack.	https://robu.in/product/0-5w-8ohm-trumpet-speaker-diameter-28mm/
Electronics	BACKUP handset cord: mic V+, mic signal, earpiece, shared ground. Old coiled cords are often	https://robu.in/product/22awg-multicore-insulated-cable4-cores-silicone/
Electronics	Flow: 'Taps on the NFC tag'. Stuck on the phone body; holds the URL of the surprise web page	https://www.amazon.in/Robodo-Sticker-13-56MHz-ISO14443A-ntag213/dp/B07DDDGTL
Electronics	Powers the board in the cab via micro-USB; driver swaps/charges it. Must NOT auto-shut-	https://www.amazon.in/dp/B0H1QCZ87F
Electronics	One for programming the board from a laptop, one from the power bank in the cab.	https://www.amazon.in/dp/B082LZGK39
Electronics	Wiring the 7 keys + hook switch to I036, I013, I019, I023, I018, I05, I021, I022 and the handset	https://www.amazon.in/dp/B0H8358SGS
Electronics	Quick connections from the board's pin header to the switch wires during testing.	https://robu.in/product/20cm-dupont-wire-color-jumper-cable-2-54mm-1p-1p-female-female-40pcs/
Electronics	In case the A1S header pins arrive unpopulated or you need extra pins to solder to.	https://robu.in/product/2-54mm-1x40-pin-male-single-row-straight-long-header-strip-pack-of-3/

Misc	Insulating every solder joint on the handset and keypad wiring.	https://www.amazon.in/dp/B08S3TP2Q6
Misc	Fixing the board, mic module and SD card inside the phone shell so nothing rattles in	https://www.amazon.in/dp/B0FD456NSD
Mechanical	Mounts the phone in the cab: phone screws to the plate, plate cable-ties to the seat frame /	https://www.amazon.in/dp/B0DPV3BKWF
Mechanical	Where the power cable exits the phone base.	https://www.amazon.in/Rebower-Rubber-Grommets-Flexible-Wire/dp/B0BN472J62

in): every link is a direct product page that was IN STOCK at t
Online orders placed by 26 Sep reach Mumbai by 29 Sep (use the

if any must be bought)

ts for the base plate

ools library and CP210x USB driver installed and tested BEFORE
to the tags)

urchase needed)

ual, 1 file) + for EACH of en/hi/mr: desc, menu, heres, none_ye
rprise, goodbye, plus beep.wav = 32 files

/hi, archive/mr so the first listener never hears 'no messages

cab location + time); free hosting (GitHub Pages / Netlify); wr

language -> description -> listen / record -> 6 / 7 -> save ->
hm resistor across 5V/GND, then leave it running on the bank ov

| (pre-loaded) and spare mic module

Additional details	Cost
LOCAL: Chor Bazaar (Mutton St) or OLX Mumbai. Test hook switch, keypad and earpiece before buying. Buy by 28 Sep.	Rs 2,500
Robu pack of 10, Rs 49, in stock (checked 25 Sep).	Rs 49

Robu KW1-103 short-lever, Rs 14 Rs 28
each, in stock (checked 25 Sep).
Order 2.

In stock at Probots, Rs 2,499 each, Rs 4,998
dispatch 1 working day, 1-3 day
express to Mumbai. Alt: Amazon.in
'ESP32-Audio-Kit ESP32-A1S'
(Rakstore / JESSINIE). ORDER BY 26

Amazon.in HP 32 GB Class 10 U1, Rs Rs 899
899, in stock (checked 25 Sep). Card
prices are high online right now - a
local mobile shop is ~Rs 400-500;
buy the 2nd (spare) card there if

Amazon.in Digitek DCR-007, USB-A 3.0 Rs 299
+ Type-C, Rs 299, in stock (checked
25 Sep).

Robu Rs 191 each, in stock (checked Rs 382
25 Sep). Alt:
<https://www.sunrom.com/p/max9814-microphone-amplifier-agc-auto-gain-control> . Order by 26 Sep.

Amazon.in Yellow Tommy pack of 10, Rs 249
Rs 249 (team-verified listing). Same
capsule type also on Lamington Road.

Amazon.in AVS 500 pc assorted kit, Rs 199
Rs 199, in stock (checked 25 Sep).
Robu resistor kits were all out of
stock. Also cheap at Lamington Road.

Robu AUX 2 m cable, Rs 279-299, in Rs 60
stock - but overpriced. Any Rs 60-
100 aux cable from a mobile shop /
Lamington Road does the same job.
Cut in half = two solderable plugs.

Robu 28 mm 0.5 W 8 ohm, Rs 39, IN STOCK (checked 25 Sep). Buy 2: one spare earpiece, one bench test speaker. The 36 mm version is out of stock.

Rs 78

Robu 22 AWG 4-core silicone cable, Rs 87 per metre, IN STOCK (checked 25 Sep). The 26/24/30 AWG versions are out of stock.

Rs 87

Amazon.in Robodo 10-pack. Alt in stock at Robu, Rs 29 per tag (checked 25 Sep):
<https://robu.in/product/13-56-mhz-ntag213-144byte-sticker-nfc-tag/> .

Rs 300

Amazon.in EVM EnBolt 10000 mAh, Rs 749, in stock (checked 25 Sep). No low-current mode - so firmware keeps Wi-Fi on + a 47 ohm 1 W keep-alive resistor across 5V/GND (from the

Rs 749

Amazon.in Ambrane 3A braided micro-USB DATA cable 1.5 m, Rs 149, in stock (checked 25 Sep). Order 2 (one for programming, one for the cab).

Rs 298

Amazon.in SixOhms 22 AWG solid-core kit, 5 colours x 2 m, Rs 99, in stock (checked 25 Sep). Robu hook-up wire kits are out of stock / Rs 800+. Lamington Road sells by the

Rs 99

Robu 20 cm F-F 40 pcs, Rs 44, in stock (checked 25 Sep). The M-M/M-F/F-F combo pack is out of stock.

Rs 44

Robu pack of 3, Rs 15, in stock (checked 25 Sep).

Rs 15

Amazon.in Rpi Shop 80 pc assorted kit, Rs 165, in stock (checked 25 Sep). Robu alt in stock:

<https://robu.in/product/colorful-heat-shrink-tubing-hst-insulation->

Amazon.in ProKart 7 mm glue sticks, 10 pcs, Rs 72, in stock (checked 25 Sep). Add double-sided foam tape from any stationery shop (~Rs 60). Rs 132

Amazon.in black acrylic 200 x 200 x 3 mm, Rs 203, in stock (checked 25 Sep). M4 bolts/nuts + cable ties from any hardware shop (~Rs 100). Pre-drill before Goa. Rs 303

Amazon.in Rebower 8 mm grommets, 20 pcs. Any hardware shop also has these. Rs 20

TOTAL	Rs 11,953
Budget	Rs 12,000
Remaining	Rs 47

he time of checking. Phone price depends on Mumbai pincode at checkout, not Goa). Nothing

leaving

t, rec_intro ('record your message in

yet'

ite its URL with cab number to the NFC tags

hang-up resets to idle) before packing -
ernight to prove the bank does not switch off